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CS499

Artifact 3: Animal Shelter

Databases

Description of the Artifact

The third artifact I selected for my ePortfolio is my Animal Shelter project from CS340. This project used Python and MongoDB to work with animal shelter information.

One of the main parts of the project was an AnimalShelter class that provided a way for the Python program to communicate with the MongoDB database. The project gave me experience with database operations and working with stored information through code.

Why I Selected This Artifact

I selected this project because databases are an important part of many software applications. A program needs a way to store and retrieve information, and MongoDB gave me experience working with a NoSQL database.

The original project demonstrated basic database functionality, but I wanted to make the searching functionality more useful. For the enhancement, I added reusable search options that allow the program to search using information such as animal type, breed, age range, and combinations of filters.

This improvement made the database class more flexible because it could be used for different searches instead of relying on one basic type of query.

What I Learned From the Enhancement

The enhancement helped me understand how important it is to make database code reusable. If a program has to perform different searches, it is better to have a flexible way to provide search information instead of creating completely separate code for every possible search.

I also learned more about how Python can be used to communicate with MongoDB. The Python code acts as the connection between the user or application and the database.

Another thing I learned was that database design affects how useful an application can be. If the information can be searched and filtered in different ways, the application can provide more useful results.

Challenges and Feedback

One challenge with the project was working with MongoDB and making sure the Python code was communicating with the database correctly. I had to troubleshoot database connection and authentication issues during my coursework.

The CS499 enhancement process gave me another chance to look at the original database code and think about how it could be made more useful. Instead of completely replacing the original functionality, I kept the existing CRUD structure and added additional searching options.

This helped me understand that an enhancement does not always mean replacing something that already works. Sometimes the better solution is to keep the existing functionality and build on top of it.

Course Outcomes

This artifact demonstrates my ability to use computing tools and techniques to implement a solution involving a database. It also demonstrates my ability to evaluate an existing solution and improve its functionality.

The project connects directly to the database portion of my computer science education and gives employers an example of my experience working with Python and MongoDB.